

REPLIT PROMPT — YachtWorth: Charter Planner

Module

CONTEXT

This is a new screen inside the existing YachtWorth app (React Native / Expo). The app already has: bottom tab navigation, Supabase backend, existing design system. This module plugs into the "Charter" entry point already created in the Tools tab. Do NOT change any existing screens. Build only what is described here.

OVERVIEW

Build a complete Charter Planner tool with 4 main views:

1. Fleet View — list of all managed yachts
2. Master Calendar — all yachts on one timeline
3. Yacht Calendar — single yacht full year view
4. Charter Form — full booking/trip data entry

All data stored in Supabase. All views are interconnected.

NAVIGATION STRUCTURE

```
CharterPlannerScreen (entry from Tools tab)
├── FleetView (default tab)
├── MasterCalendarView
├── YachtDetailScreen (from fleet card tap)
│   ├── YachtCalendarView
│   └── CharterFormScreen (from calendar tap)
│       └── CharterPLScreen (P&L summary, auto-calculated)
```

Top tab bar inside Charter Planner:

[Fleet] [Calendar]

Both tabs always visible. Switching is instant.

SUPABASE SCHEMA

Create these tables if they don't exist:

```
-- Yachts fleet
create table yachts (
  id uuid primary key default gen_random_uuid(),
  user_id text not null,
  name text not null,
  type text, -- 'Motor Yacht', 'Sailing', 'Catamaran', 'Superyacht'
  length_m numeric,
  year integer,
  flag text,
  home_port text,
  photo_url text,
  notes text,
  created_at timestamptz default now()
);

-- Charters / bookings
create table charters (
  id uuid primary key default gen_random_uuid(),
  yacht_id uuid references yachts(id) on delete cascade,
  user_id text not null,
  status text default 'confirmed', -- confirmed | tentative | maintenance | block
  client_name text,
  client_email text,
  client_phone text,
  start_date date not null,
  end_date date not null,
  departure_port text,
  return_port text,
  -- Vessel
  engine_hours_before numeric,
  engine_hours_after numeric,
  fuel_liters numeric,
  fuel_price_per_liter numeric,
  -- Crew
  captain_name text,
  captain_day_rate numeric,
```

```

    stewardess_count integer default 0,
    stewardess_day_rate numeric,
    extra_crew_cost numeric,
    -- Revenue
    charter_rate_type text default 'fixed', -- 'fixed' | 'per_day'
    charter_rate numeric,
    deposit_amount numeric,
    deposit_date date,
    final_payment_amount numeric,
    final_payment_date date,
    vat_applicable boolean default false,
    vat_percent numeric default 0,
    -- Expenses
    port_fees numeric default 0,
    provisioning numeric default 0,
    cleaning numeric default 0,
    other_expenses numeric default 0,
    other_expenses_note text,
    -- Notes
    notes text,
    created_at timestamptz default now(),
    updated_at timestamptz default now()
);

-- Clients (auto-populated from charters)
create table clients (
    id uuid primary key default gen_random_uuid(),
    user_id text not null,
    name text not null,
    email text,
    phone text,
    notes text,
    created_at timestamptz default now()
);

```

VIEW 1: FLEET VIEW

Header:

My Fleet

[+ Add Yacht]

Empty state:

[anchor icon, large, gold]
No yachts in your fleet yet.
Add your first yacht to start planning.

[+ Add my first yacht]

Yacht card (when yachts exist):

[photo or type icon]	Aurelia	
	Azimut · 24m · 2019	
	Malta flag	
STATUS TODAY:	● Available	
Next charter:	Jun 15 - Jun 22	
This month:	3 charters · 67% booked	

Status colors:

- Available → green
- Chartered → gold
- Maintenance → grey
- Option → blue

Tap card → opens YachtDetailScreen **Swipe left** → shows [Edit] [Archive] buttons **Long press** → context menu: Edit · Duplicate · Archive · Delete

Add Yacht bottom sheet / modal: Fields:

- Name (required)
- Type: pill selector [Motor] [Sailing] [Catamaran] [Superyacht]
- Length (m)
- Year built
- Flag / registration
- Home port
- Notes
- [Save yacht] gold button

VIEW 2: MASTER CALENDAR

Layout: Gantt-style grid

	1	2	3	4	5	6	7	...	30	31
Aurelia										
Lady M										
Blue One										

- Each row = one yacht
- Each column = one day
- Color blocks = charter status colors
- Horizontal scroll for days
- Vertical scroll for yachts (if more than 5)

Month navigation:

< June 2026 >

Arrow buttons change month. Current day highlighted with gold border.

Tap on a day column header → Day Detail Panel slides up:

June 15, 2026

- Aurelia CHARTERED →
- Lady M Available + Add
- Blue One OPTION →

Each row tappable:

- CHARTERED / OPTION → opens that charter form
- Available + Add → opens new charter form with that yacht pre-selected and date pre-filled

Tap on a colored block → opens that charter form directly

Legend at top right: Small colored dots with labels: Confirmed · Tentative · Maintenance · Blocked

VIEW 3: YACHT DETAIL SCREEN

Header: [← back] Aurelia · Azimut 24m

Two sub-tabs:

[Calendar] [Info]

Calendar sub-tab:

Full year view of this single yacht. 12 months displayed vertically (scrollable). Each month shows day grid with color blocks for charters.

Tap occupied day → opens that charter form **Tap empty day** → bottom sheet:

June 15 — Aurelia

[+ New Charter]

[+ Block dates]

[+ Mark maintenance]

Info sub-tab:

Shows yacht details (editable). Summary stats:

- Total charters this year
- Total revenue this year (€)
- Total booked days / available days
- Average charter duration (days)
- Most frequent client

VIEW 4: CHARTER FORM SCREEN

Header: [← back] [New Charter] or [Edit Charter] Progress indicator: ●●●●●●● (7 sections, scroll-through, not steps)

All sections on one scrollable screen. Section headers are collapsible.

Section 1: BASICS

CLIENT NAME *

[_____]

CLIENT EMAIL

[_____]

CLIENT PHONE

[_____]

STATUS

[Confirmed] [Tentative] [Maintenance] [Blocked] [Cancelled]

DATES *

Start: [17 Jun 15, 2026] End: [17 Jun 22, 2026]

Duration: 7 days ← auto-calculated, shown in gold

DEPARTURE PORT

[_____]

RETURN PORT

[_____]

Section 2: VESSEL

YACHT

[Aurelia – Azimut 24m] ← pre-selected, tappable to change

ENGINE HOURS BEFORE DEPARTURE

[_____] hrs

ENGINE HOURS AFTER RETURN

[_____] hrs

Engine hours used: X hrs ← auto-calculated

FUEL CONSUMED

[_____] liters

FUEL PRICE PER LITER

[_____] €

Fuel cost: €X,XXX ← auto-calculated in gold

Section 3: CREW

CAPTAIN

[Name: _____] [Day rate: ____€/day]

Captain total: €X,XXX ← auto (rate × days)

STEWARDESS

Count: [-] 1 [+] Day rate: [____€/day]

Stewardess total: €X,XXX ← auto

ADDITIONAL CREW / EXTRAS

[_____] €

[Description: _____]

TOTAL CREW COST: €X,XXX ← bold, auto-calculated

Section 4: REVENUE

CHARTER RATE TYPE

[Fixed total] [Per day]

CHARTER RATE

[_____] €

← if per day: shows "× 7 days = €X,XXX total"

DEPOSIT

Amount: [_____] €

Date received: [ 17]

Status: [Received ✓] [Pending]

FINAL PAYMENT

Amount: [_____] €

Date: [ 17]

Status: [Received ✓] [Pending]

VAT

[Apply VAT] toggle

VAT %: [_____]%

GROSS REVENUE: €X,XXX

VAT AMOUNT: €X,XXX

NET REVENUE: €X,XXX ← gold, prominent

Section 5: EXPENSES

PORT FEES	[_____]	€
PROVISIONING	[_____]	€
CLEANING	[_____]	€
OTHER EXPENSES	[_____]	€
Description:	[_____]	

TOTAL EXPENSES:	€X,XXX	← auto
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Section 6: P&L SUMMARY

Auto-calculated. Read-only display. Updates live as user fills form.

P&L THIS CHARTER			
Net Revenue	€X,XXX		
- Crew costs	€X,XXX		
- Fuel	€X,XXX		
- Expenses	€X,XXX		
NET PROFIT	€X,XXX	← green if positive, red if negative	
MARGIN	XX%		

Section 7: NOTES

NOTES

[_____]

[_____]

[Multiline free text field]

Bottom action bar (sticky):

[ Delete] [Export PDF] [Export XLS] [Save Charter ✓]

Save → validates required fields → saves to Supabase → returns to calendar with new block visible.

REPORTS & EXPORTS

PDF Report — Single Charter:

Generate clean PDF with:

- Yacht name + dates + client
- P&L table
- All filled fields
- Footer: "Generated by YachtWorth · Powered by PDYE"

PDF Report — Monthly Fleet Summary:

For each yacht:

- Number of charters
- Total booked days vs available
- Total gross revenue
- Total expenses
- Net profit
- Occupancy % (booked days / total days in month)

Footer totals across all yachts.

Excel Export — Charter Log:

One row per charter, all fields as columns. Compatible with standard spreadsheet apps.

CLIENT HISTORY (accessible from Fleet View → Clients tab)

Add third tab to Charter Planner top nav:

[Fleet] [Calendar] [Clients]

Client list — alphabetical or by most recent. Each client card shows:

- Name
- Total charters: X
- Total spent: €X,XXX
- Last charter: date

Tap client → see all their charters chronologically. Search bar at top. Clients auto-created when charter is saved with a client name.

TECHNICAL REQUIREMENTS

1. All monetary values formatted: €1,234.56 — no exceptions
2. All P&L calculations are real-time — update as user types
3. Date pickers use native DateTimePicker
4. Calendar grid: custom component — do NOT use a calendar library that doesn't support Gantt-style multi-row layout
5. Supabase: use existing Supabase client already configured in the app
6. user_id: use existing auth session user ID
7. Offline: show "Saving..." indicator, queue writes if offline
8. Colors: use existing tokens — #0D1B2A , #C5973A , #1A2B3C , #FFFFFF , #8899AA
9. Charter status colors:
 - confirmed: #2ECC71 (green)
 - tentative: #3498DB (blue)
 - maintenance: #95A5A6 (grey)
 - blocked: #E74C3C (red)
 - cancelled: #7F8C8D (dark grey, strikethrough)
10. PDF generation: use react-native-html-to-pdf or expo-print

11. Excel export: use SheetJS (xlsx library)
 12. All delete actions require confirmation dialog
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FILE STRUCTURE

```
/screens/Charter/
  CharterPlannerScreen.js    ← entry, top tab navigator
  FleetView.js              ← yacht list
  MasterCalendarView.js     ← gantt grid all yachts
  YachtDetailScreen.js      ← single yacht detail + calendar
  CharterFormScreen.js      ← full charter data entry
  ClientsView.js            ← client history list
  ClientDetailScreen.js     ← single client all charters

/components/Charter/
  YachtCard.js              ← fleet list card
  GanttGrid.js              ← master calendar grid component
  MonthCalendar.js          ← single yacht month view
  CharterBlock.js           ← colored block in calendar
  DayDetailPanel.js         ← bottom sheet for day tap
  PLSummaryCard.js          ← P&L auto-calc display
  AddYachtSheet.js          ← bottom sheet to add yacht
  CharterStatusBadge.js     ← colored status pill

/utils/Charter/
  charterCalculations.js    ← all P&L formulas (pure functions)
  exportPDF.js              ← PDF generation logic
  exportExcel.js            ← Excel export logic
  supabaseCharter.js        ← all Supabase queries for this module

/hooks/
  useFleet.js               ← fetch and manage yacht fleet
  useCharters.js            ← fetch charters by yacht/date range
  useClients.js             ← fetch client history
```

PRIORITY BUILD ORDER

1. Supabase tables (create schema)
2. `supabaseCharter.js` — all CRUD operations

3. `charterCalculations.js` — all P&L formulas
 4. `FleetView.js` — yacht list + add yacht
 5. `CharterFormScreen.js` — full form (most complex, build first)
 6. `MonthCalendar.js` component — single yacht view
 7. `YachtDetailScreen.js` — wraps calendar
 8. `GanttGrid.js` — master calendar (most complex UI)
 9. `MasterCalendarView.js` — wraps gantt
 10. `ClientsView.js` + `ClientDetailScreen.js`
 11. Export functions (PDF + Excel)
 12. `CharterPlannerScreen.js` — top tab navigator connecting all
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IMPORTANT NOTES FOR DEVELOPER

- The yacht objects in this module should share the same `yacht_id` as yachts stored elsewhere in the app (valuation, cost estimator). If a yacht already exists in the app's yacht table, it should appear in the fleet automatically — do not create duplicate records.
 - The charter form is the core of this module — invest the most time here. Every field must save correctly and every calculation must be accurate.
 - The Gantt calendar is visually the most complex — if performance is an issue with many yachts and many months, use `FlatList` with horizontal + vertical scroll and memoized row components.
 - Do not build the Digital Passport integration yet — just make sure `yacht_id` is consistently used so the connection can be made later.
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